Summary in Graph

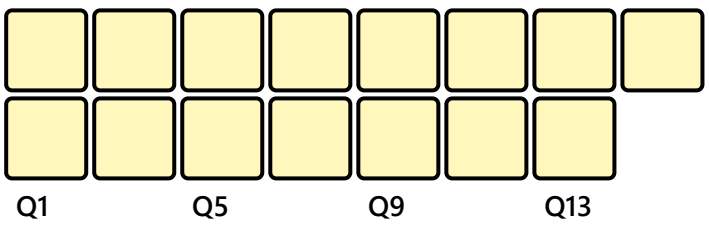
Exam Summary (GO Classes CS | CO and Architecture | Cache Memory | Weekly Quiz 5)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (0 + 15)
Total Marks:	30 (0 + 30)
Exam Duration:	60 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (14) out of 141 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 2 Penalty: 0.67 CO & Architecture

A cache memory needs an access time of 30 ns and main memory 150 ns, what is the average access time of the CPU (assume hit ratio $= 80\%$)?

A. 60 ns

B. 30 ns

C. 150 ns

D. 70 ns

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

The L3 cache of a computer has a hit time of 15 cycles and a miss penalty of 300 cycles. Suppose that the hit rate for this cache, for a given application, is 95%. Considering just this cache, what is the average memory access time in cycles?

Your Answer: | Correct Answer: 30 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

The timing for a particular cache is as follows: checking the cache takes 1 cycle. If there's a hit the data is returned to the CPU at the end of the first cycle. If there's a miss, it takes 10 additional cycles to retrieve the word from main memory, store it in the cache, and return it to the CPU. If we want an average memory access time of 1.4 cycles, what is the minimum possible value for the cache's hit ratio (in $\%$)?

Your Answer: | Correct Answer: 96 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

The Beta Engineering Team is working on the design of a cache.
This cache takes 2 clock cycles to determine if a memory access is a hit or a miss and, if it's a hit, return data to the Beta(i.e. the CPU). If the access is a miss, the cache takes 20 additional clock cycles to fill the cache line and return the requested word to the Beta. If the hit rate is 90%, what is the Beta's average memory access time in clock cycles?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a 2-way set-associative cache where each way has 4 cache lines with a block size of 2 words. Using this cache, a particular benchmark program experiences an average memory access time (AMAT) of 1.3 cycles. The access time on a cache hit is 1 cycle; the miss penalty (i.e., additional access time) is 10 cycles. What is the hit ratio when running the benchmark program (in $\%$)?

Your Answer: | Correct Answer: 97 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Find the AMAT for a processor with a 1 ns clock cycle time, a miss penalty of 20 clock cycles, a miss rate of 0.05 misses per instruction, and a cache access time (including hit detection) of 1 clock cycle.
Assume that the read and write miss penalties are the same and ignore other write stalls.

The average memory access time (in ns) per instruction is?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is _____.

Your Answer: | Correct Answer: 14 | Not Attempted | Time taken: 00min 01sec | Discuss

Q #8 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider that a computer system has 5 ns of latency when accessing cache and 70 ns when accessing main memory. What should your application's average hit rate be (in $\%$) in order to have an average memory access latency of 15 ns (Round to two decimal places)?

Your Answer: | Correct Answer: 85.71 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Assume that checking the cache on each read takes 1 cycle and that refilling the cache on a miss takes an additional 8 cycles. If we wanted the average access time over many reads to be 1.1 cycles, what is the minimum hit ratio the cache must achieve during that period of time (in $\%$) ?

Your Answer: | Correct Answer: 98.75 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A cache memory that has a hit rate of 0.8 has an access latency 10 ns and miss penalty 100 ns. An optimization is done on the cache to reduce the miss rate. However, the optimization results in an increase of cache access latency to 15 ns, whereas the miss penalty is not affected. The minimum hit rate *(rounded off to two decimal places)* needed after the optimization such that it should not increase the average memory access time is _____.

Your Answer: | Correct Answer: 0.85 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A computer has a single cache (off-chip) with a 3 ns hit time and a 95% hit rate. Main memory has a 50 ns access time. If we add an on-chip cache with a 0.6 ns hit time and a 98% hit rate, the computer's effective access time:

A. 2.8 ns

B. 5.5 ns

C. 0.7 ns

D. None of the options

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

In a two-level cache system, the access times of the L_1 and L_2 caches are 1 and 8 clock cycles, respectively. The miss penalty from the L_2 cache to main memory is 18 clock cycles. The miss rate of the L_1 cache is twice that of L_2. The average memory access time (AMAT) of this cache system is 2 cycles. The miss rates of L_1 and L_2 respectively are

A. 0.111 and 0.056

B. 0.056 and 0.111

C. 0.0892 and 0.1784

D. 0.1784 and 0.0892

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

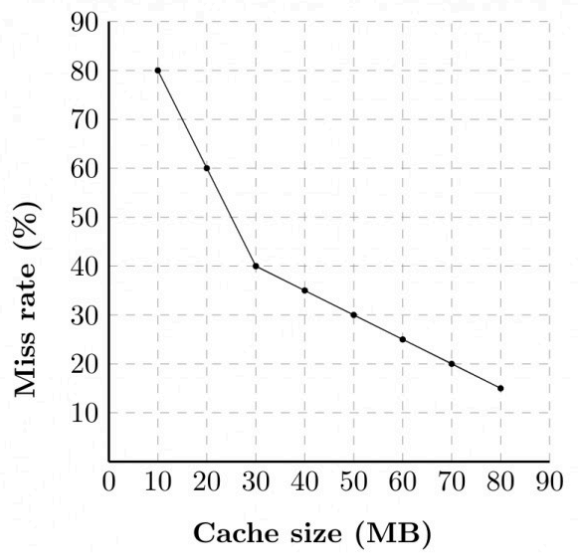
Q #13 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a memory system with a cache access time of 10 ns and a memory access time of 110 ns assume the memory access time includes the time to check the cache. If the effective access time is 10% greater than the cache access time, what is the hit ratio H? (fractional answer)

Your Answer: | Correct Answer: 0.99 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A file system uses an in-memory cache to cache disk blocks. The miss rate of the cache is shown in the figure. The latency to read a block from the cache is 1 ms and to read a block from the disk is 10 ms. Assume that the cost of checking whether a block exists in the cache is negligible. Available cache sizes are in multiples of 10 MB.



The smallest cache size required to ensure an average read latency of less than 6 ms is _____ MB.

Your Answer: | Correct Answer: 30 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a system with a 2 level cache. Access times of Level 1 cache, Level 2 cache and main memory are 1 ns, 10 ns, and 500 ns, respectively. The hit rates of Level 1 and Level 2 caches are 0.8 and 0.9, respectively. What is the average access time of the system, ignoring the search time within the cache?

A. 13.0

B. 12.8

C. 12.6

D. 12.4

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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